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Assessing the potato yield gap in the Peruvian Central Andes

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ABSTRACT

The Peruvian Central Andes is a highly important area for potato production. Assessing the potato yield gap and the potential yield is an essential step towards sustainable crop intensification. Fifty-eight smallholder potato farmer's plots in total were monitored at field level during the 2005–2008 and 2010–2015 rainy cropping seasons. All the main crop management inputs were registered. Three field experiments (on-farm trials) established during the 2014–2017 rainy cropping seasons were used to calibrate (2014–2016) and validate (2016–2017) the SUBSTOR-potato model under potential conditions. Potential potato yield (Y_p) was estimated for each individual field pilot plot (in kg ha^{-1}) based on the calibrated and validated crop model. Yield gaps (Y_g) were calculated as the difference between Y_p and farmers' actual yield (Y_a). A classification tree-based model predicting the potato gap quantiles was used to elucidate the main biophysical and crop management components inducing Y_g.

Performance of the SUBSTOR-potato model showed a close agreement of simulated crop biomass, tuber yield, and N-uptake (i.e. N-demand) with the measured data under potential conditions. Redefined index of agreement were 0.84 and 0.80 while the associated mean square error were 2232 and 916 kg tuber dry weight (DW) ha^{-1} for the calibration and validation, respectively. The mean farmers' actual DW yield was 7118 kg ha^{-1}, however, a high variability due to heterogeneous biophysical conditions and crop management was found (from 710 to 18,885 kg DW ha^{-1}). The potato Y_g ranged from 0.1 to 95.8% of the potential yield ($\bar{x}$ = 42.1%, $\tilde{x}$ = 46.0%, σ_x = 28.14% and CV = 0.67), hence there is an important difference that needs to be reduced. The classification tree analysis showed that inorganic N is the main yield explaining factor. While large yield gaps (Fourth quantile) are induced by low Inorganic N (< 88 kg ha^{-1}) and scarce Human Labour energy (< 4196 MJ ha^{-1}), small yield gap (First quantile) is mainly attributed to high N-inputs (≥ 139 kg ha^{-1} inorganic and ≥ 154 kg ha^{-1} organic). Third and Second quantiles (mid potato yield gaps) were characterized by more intricate nutrient input use, being difficult to classify; the Third quantile was partially explained by Inorganic N (< 139 kg ha^{-1}), while part of the Second quantile by Extractable soil phosphorus (< 7.3 mg kg^{-1}) and Inorganic N (< 139 kg ha^{-1}). This classification can be helpful to diagnose the main site-specific crop management and biophysical recommendations towards closing the potato yield gap.

The analysis suggests that there is opportunity to enhance potato actual yields in the study zone. More rational amount of inputs together with best management practices might improve potato productivities. However, sustainable potato intensification should be complemented with the expected quantification of environmental burdens under the local socio-economic constraints.

1. Introduction

Potato is the third most important cultivated food crop in the world (FAOSTAT, 2018), with an average annual production of 356 million of tons during the last decade (2007–2016) and is believed to contribute significantly to sustain future global food security (Burlingame et al., 2009). Its cultivation is a main resource for employment, income, and food security in developing countries (Lutaladio and Castaldi, 2009). Total productions for developed and developing countries have shown similar trends (Devaux et al., 2014). However, developing countries nowadays produce a larger volume than the developed countries (Haverkort and Struik, 2015).

One of the most relevant options for sustainable intensification of agricultural systems, especially in developing countries, is to bridge the

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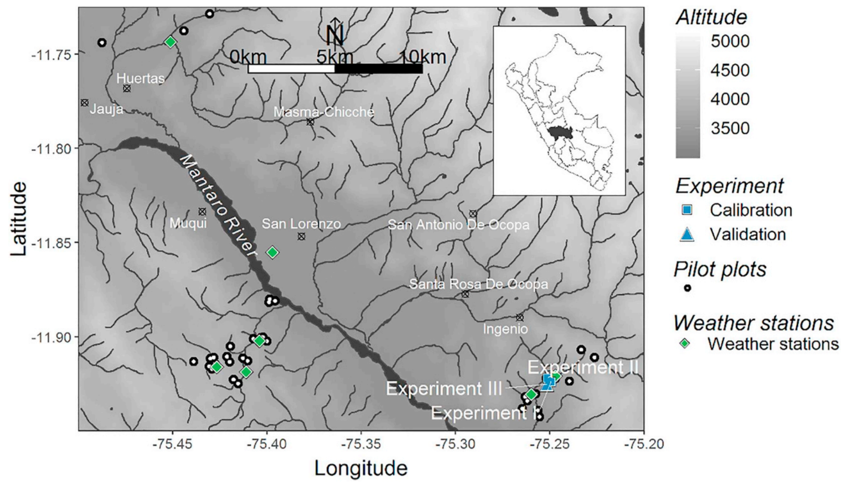

Fig. 1. Location of the study zone and monitored potato farms in the Mantaro Valley, Junin, Peru. Square: calibration experiments. Triangle: validation experiment. Circle: potato farmer's plots. Rhombus: weather stations.

yield gap (Y_g) (Cassman et al., 2010; van Ittersum et al., 2013). While Y_g analysis contributes to identify relevant crop, soil, and management factors limiting current farm yield (van Ittersum and Rabbinge, 1997; van Ittersum et al., 2013), Y_g provides a quantitative estimate of the possible increase in crop productivity (van Wart et al., 2013; van Oort et al., 2017). Nevertheless, for a deeper understanding of Y_g, more local studies are needed to investigate the role of farm(er) characteristics considering the local biophysical and socioeconomic conditions (Beza et al., 2017).

Estimating Y_g at crop level requires comparing farmers' actual yield (Y_a) to a reference of potential yield (Y_p). Y_a is the actually achieved yield in a farmer's field. To represent variation in time and space in a defined geographical region, Y_a is defined by the most widely used management practices related to the availability of inputs, climate variability, technology level, and crop management (van Ittersum et al., 2013). On the other hand, Y_p is the yield obtained without water, nutrient nor biotic stress (Evans, 1993; van Ittersum and Rabbinge, 1997). Y_p is location-specific because of the climate, but in theory not dependent on soil properties assuming that the required water and nutrients can be added through management (van Ittersum et al., 2013). Under these optimal conditions, crop growth is determined by solar radiation, temperature, atmospheric CO_2 concentration, crop genetic traits, and management practices (Rattalino Edreira et al., 2017). The measure of Y_p, when compared to Y_a, allows the calculation of the Y_g (Lobell et al., 2009). The approach to assess the magnitude and causes of Y_g in specific areas involves conducting controlled research trials to evaluate various input levels or management practices eliminating yield-limiting factors (Rattalino Edreira et al., 2017). This approach in combination with a well-calibrated and -validated crop simulation model, has been shown to be an important approach for local studies (Lobell et al., 2009; van Ittersum et al., 2013; Grassini et al., 2015). As such, crop models provide the means to capture spatial and temporal variation, to the extent that climate data are location-specific (van Ittersum et al., 2013).

While potato production in the Andes has been explored under cultivar variations and different growing conditions using the SUBSTOR-potato model (Kleinwechter et al., 2016; Raymundo et al., 2017), limited research has been developed to evaluate the Y_g. Potato is the most important temporary cash and staple crop in Peru (Tobin et al., 2016; INEI, 2012), with a national average fresh weight yield of 14.2 t ha^{-1} (FAOSTAT, 2018). It is cultivated predominantly by smallholders in the Andes under heterogeneous biophysical conditions and crop management practices during the rainy growing seasons when there is no water limitation (García, 2011). During the last decades, Peruvian Andean potato farmers have been adapting classical low-yielding technology (e.g. animal traction, human labour, and scarce inputs) to local constraints (Escobal and Cavero, 2012). These practices have resulted in a relatively good site-specific management of sub-processes (e.g. tillage, seeding, and weeding) (García, 2011). Nonetheless, overall optimization of the production systems requires further improvements related to close the Y_g and best management strategies for sustainable intensification of these agro-ecosystems. Therefore, quantifying the Y_g is crucial to estimate the achievable degree of yield improvement for potato crops in the Andes. This study was performed with the main objective to assess the Y_g of smallholder potato production in the Peruvian Central Andes. The specific objectives of the present study were to (a) calibrate and validate the SUBSTOR-potato model under local potential conditions, (b) estimate the potential potato yield, and (c) assess the main biophysical and crop management components to elucidate the potato yield gap.

2. Material and methods

2.1. Study area and climate conditions

The study zone was located in the Mantaro Valley, Central Andes of Peru (Fig. 1), one of the country's main centers of potato production (Escobal and Cavero, 2012). This region belongs to the Tropical Highlands potato mega-environment (You et al., 2014; Raymundo et al., 2018). Potato is cultivated during the rainy season, from October/November to April/May. Seven full weather stations were installed along the study zone to describe the local climate during the research period from 2005 until 2017 (Fig. 1). Climate data was collected hourly and the meteorological equipment was calibrated yearly. A preliminary data quality assessment was performed following the recommendations of Hunziker et al. (2017). A general overview of the climate conditions (on monthly basis) is depicted in Fig. 2. The mean air temperature over the different stations fluctuated from 8.8 to 11.6 °C, with an average value of 10.6–11.6 °C during the growing season (Fig. 2A). Incoming solar radiation ranged from 470 to 589 MJ m^{-2} month^{-1}, reaching a total value of 3211–3709 MJ m^{-2} on average during the cropping season (Fig. 2B). Average total rainfall was 833 mm year^{-1}, of which approximately 83–89% fell during the growing season (Fig. 2C). Potential evapotranspiration (ASCE-EWRI, 2005) ranged 677–699 mm on average during the cropping seasons (Fig. 2D).

2.2. Field pilot plots - Farmers' yields

Field data were collected in the cropping seasons 2005–2008 and 2010–2015. Fifty-eight potato farmers' plots located in the study zone

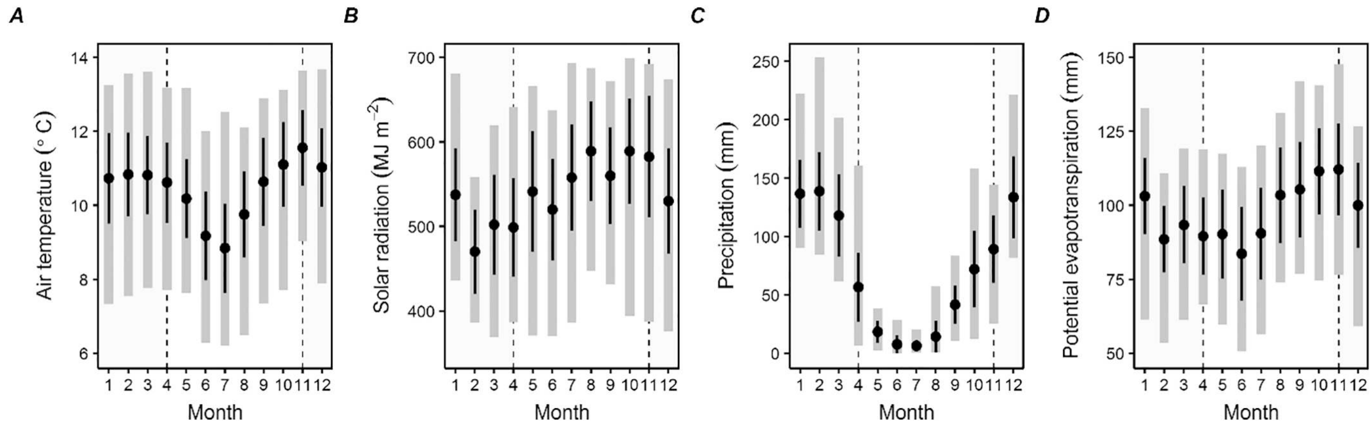

Fig. 2. Average monthly (A) air temperature (°C), (B) solar radiation (MJ m^{-2}), (C) precipitation (mm) and (D) potential evapotranspiration (mm) between 2005 and 2017 in the study zone. The black point represents the mean, while the black line and grey bar show ± standard deviation and maximum-minimum, respectively. Shaded light grey area shows the start and end of the cropping seasons.

were evaluated at field level (Fig. 1). These field pilot plots ($N=58$) were observed in a transect of the Mantaro Valley, ranging from 3312 to 4014 m above sea level (m.a.s.l.). Detailed follow-ups through fortnightly basis visits of all the pilot plots was performed; therefore, all the cultivation technologies were measured on the field. For each specific farmer-cropping season, fresh and dry yield, aboveground residual biomass, and weeds (kg) were measured on subplots with known planting density (in plant m^{-2} measured on 3 rows of 10-m long approximately) taking three replicates per plot and scaling-up for the plot size at harvest (kg ha^{-1}). The subplots were random spatially determined within the plot. Additionally, main crop management inputs were registered per plot: inorganic and organic fertilization (kg ha^{-1}), human and animal labor (h ha^{-1}), use of machinery (h ha^{-1}), pesticides and fungicides (kg ha^{-1}). Nutrient chemical analysis was performed for the organic fertilization (mainly farmyard manure) and above ground residual biomass incorporated in the fields. Additionally, composite samples were obtained from the furrow slice (0.20 m) before the cropping seasons for each field pilot plot to evaluate soil chemical properties: pH, organic carbon, total nitrogen, extractable phosphorus, and extractable potassium. Specific number of monitored field pilot plots per cropping season is provided in the supplementary material (Table S6).

2.3. Field experiments to obtain potential yield

The on-farm experiments were established in farmers' plots during the 2014–2017 cropping seasons; their location is shown in Fig. 1. The experiments were performed to test optimal nitrogen treatments to obtain the potential potato yield (i.e. ~ 300 kg N ha^{-1}). Potato cultivar *Yungay*, a popular old improved late-season variety (Urrea-Hernandez et al., 2016), was used. Experiments were performed adapting the best management practices (Zebarth and Rosen, 2007) to the local conditions: soil-based test of soil nutrient supply, manure application, split fertilizer application, and fertilizer banding. N-fertilizer was manually fractionated at planting, emergence (first ridging) and tuber initiation (second ridging). Phosphorus (P_2O_5), potassium (K_2O), calcium (CaO) and magnesium (MgO) were applied at planting and emergence at optimal doses based on preliminary soil fertility analysis and following the recommendations of Roy et al. (2006). Farmyard manure was manually applied along to best management practices at soil tillage. During the growth cycles, the fields were weeded manually; pests and diseases were controlled along to conventional practices to assure potential conditions were obtained. The specific characteristics of the on-farm experiments can be found in the supplementary material Tables S1–S3.

To assess N-uptake (i.e. N-demand under potential conditions) and biomass accumulation, five plants were selected at five critical growth and phenological stages (Zebarth and Rosen, 2007): sprout development, plant establishment, tuber initiation, tuber bulking, and tuber maturity. Sampled plants were randomly obtained from central rows to avoid border effect errors. Above- and below-ground plant biomasses were separated for chemical analysis (total nitrogen). Similarly, digital images of leaves per plant were taken on each evaluation date to obtain the leaf area index (LAI), i.e. the total leaf area per unit ground area. The results of the on-farm experiments can be found in the supplementary material Fig. S4.

For each individual experiment, the N-treatment which obtained the highest yield was selected to define the potato potential production; the N-supply where the agronomic response curve of total yield versus N reached saturation (Vos, 2009). These results were used in the following methodological procedures of Section 2.4. Specific details and additional results of the on-farm experiments are given in the supplementary material (Tables S1–S3, and Fig. S4).

2.4. Potato growth model calibration and validation

2.4.1. SUBSTOR-potato model

SUBSTOR-potato is a process-based crop model that simulates the daily crop growth and development of a potato plant (Hoogenboom et al., 2018; Jones et al., 2003). Five phenological stages are assumed for simulation: pre-planting, sprout elongation, emergence, tuber initiation, and tuber maturity. When emergence date is available, pre-planting and sprout elongation stages can be omitted. Five specific-cultivar parameters define the potato growth and development: sensitivity to temperature for tuber initiation (TC, °C), sensitivity to photoperiod for tuber initiation (P2, dimensionless), determinacy (PD, dimensionless), potential tuber growth rate (G3, g m^{-2} d^{-1}) and potential leaf area expansion (G2, cm^2 m^{-2} d^{-1}). Dry matter accumulation depends on intercepted radiation, temperature, and nutrient availability (nitrogen). Dry matter partitioning to stems and roots is based on leaf growth within the particular phenological stage. For further details of the SUBSTOR-potato model functions see Griffin et al. (1993), Ritchie et al. (1995) and Singh et al. (1998).

2.4.1.1. Cultivar calibration. Calibration and validation of model parameters and control functions were based on the potential potato field experimental data for the *Yungay* cultivar assuming no limiting factors. The on-farm experiments 2014–2016 were used for the calibration process, whereas 2016–2017 for the validation. Similarly, the daily climate data for each growing season (2014–2017) were used. The parameters were adjusted simultaneously and iteratively to obtain the final calibration parameters, comparing simulated-observed measurements and optimizing goodness-of-fit statistics according to Kleinwechter et al. (2016). The calibrated cultivar genetic coefficients were: TC = 20 °C, P2 = 0.9, PD = 0.7, G3 = 20 g m^{-2} d^{-1} and G2 = 2000 cm^2 m^{-2} d^{-1}.

2.4.1.2. Leaf area growth and carbon partitioning. Specific leaf area was set to 295 cm^{-2} g^{-1} based on measured data to correctly simulate leaf dry weight and leaf area. Cumulative tuber initiation index increased from 20 to 30 to correctly predict the tuber initiation dates. This adaptation has been recommended for white varieties with later dates of tuber initiation (Fleisher et al., 2003). Dry matter partitioning was adapted based on allometric field measurements. From tuber initiation to harvest, assimilate partitioning parameter for roots was 15% while for stems was 95% of the leaf dry weight.

2.4.2. Model performance evaluation

Goodness-of-fit statistics were used to evaluate model performance during the calibration and validation process. These indices were quantified to assess the model error and the model fit of the simulated values. The selection of the statistical indicators followed the recommendations performed by Yang et al. (2014).

Mean absolute error (*MAE*) and the root mean square error (*RMSE*) quantified the model bias in the same units of the observed and simulated variables. Both indices are commonly used to measure the error of the variables individually (Yang et al., 2014). *MAE* is less sensitive to outliers than *RMSE* (Willmott et al., 1985), and they are expressed as:

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |S_i - O_i| \tag{1}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (S_i - O_i)^2} \tag{2}$$

Where n is the number of simulated-observed pairs, S are the simulated values, and O are the observed values. Low *MAE* and *RMSE* values indicate better simulation results.

The model efficiency was assessed based on the redefined index of agreement (d_r; Willmott et al., 2011). d_r (dimensionless) is preferred over other model fit indices since it overcomes the inflating effect of extreme values, providing a clear physical meaning (Yang et al., 2014). d_r is calculated as follows:

$$d_r = 1 - \frac{\sum_{i=1}^{n} |S_i - O_i|}{2\sum_{i=1}^{n} |O_i - \overline{O}|} \tag{3}$$

Where $\overline{O}$ is the averaged observed value. d_r is bounded by −1.0 and 1.0. $d_r \approx 1$ represents a perfect fit, while $d_r \approx -1$ indicates that the averaged observed value is a better estimate than the simulated values.

A graphical representation of model performance was elaborated based on Coucheney et al. (2015). This graphic is built on the geometrical relation between the *RMSE*, the systematic error ($RMSE_s$) and the unsystematic error ($RMSE_u$) suggested by Willmott (1981): $RMSE = RMSE_s^2 + RMSE_u^2$.

$$RMSE_s = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (\widehat{S}_i - O_i)^2} \tag{4}$$

$$RMSE_u = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (S_i - \widehat{S}_i)^2} \tag{5}$$

Where $\widehat{S}_i$ is the predicted value calculated from the linear regression of the observed versus simulated values: $\widehat{S}_i = a + bO_i$. For one specific variable of evaluation, the model performance is represented by a point whose coordinates are proportional to $RMSE_s$ and $RMSE_u$ normalized by the standard deviation of the observed values (σ_O). The distance from the origin to this point is proportional to *RMSE* normalized by σ_O; the closer the point is to the origin, the smaller the error during the simulation. The normalization with the standard deviation allows to account for the variable variation range and thus to compare performance for different variables on heterogeneous dataset (Coucheney et al., 2015). As suggested by Coucheney et al. (2015), the figure was divided by the 1:1 identity line where $RMSE_s$ (bias of the model) is predominant above the diagonal and $RMSE_u$ (dispersion of the model) predominant below the diagonal.

2.5. Yield simulations, yield gap assessment, and statistical analysis

The potential yield for each field pilot plot was calculated using the SUBSTOR-potato model, previously calibrated and validated based on the on-farm experiments. Simulations were performed assuming no water nor nutrient (N) limitations considering the local measured daily climate data to obtain the simulated-potential yield (Y_p). The closest weather station was used for each individual field pilot plot and for each specific cropping season. Potato variety, plant density, sowing date, and harvesting date were also considered. The biomass components including the potential yield (per plant) were scale-up to the area of the field pilot plot. The potato yield gap (Y_g) was defined as the difference between the Y_p and the actual farmer's yield (Y_a). Similarly, the yield gap closure index (i.e. Y_a and Y_p ratio) was calculated (Rosa et al., 2018; Haverkort and Struik, 2015; Mueller et al., 2012).

Kernel density estimations (KDE) were drawn to explore the distribution and variability of the Y_a, Y_p, and Y_g. KDE is a non-parametric density estimator, not requiring any underlying density function assumption for its calculation (Chen, 2017).

The factors explaining the potato yield gap are complex, non-linear, and interdependent. In order to assess the main crop management and biophysical limitations defining the Y_g, a classification tree was built (Kotsiantis, 2011). The Y_g values were categorized in 4-quantiles based on the KDE. Therefore, Y_g values were labeled in function of the specific quantiles range: First, Second, Third, and Fourth quantile. The classification tree constructs a set of decision rules by recursive partitioning of the observations based on the multivariate space of the explanatory variables. Considered as a nonparametric discrimination method, the algorithm attempts to find patterns in the data by decreasing the impurity (e.g. Gini index) as much as possible in each split (decision rule). The set of relevant decision rules determines a binary decision tree where each terminal node is designated by a class label, based on the probability to belong to that class. A stratified K-fold cross-validation was performed to validate and measure the error of the classification tree (Luo et al., 2016). Further specific details can be found in Kotsiantis (2011). To construct the classification tree, inorganic and organic fertilizer rates were used (kg nutrient ha^{-1}). The active ingredient (kg a.i. ha^{-1}) were considered for pesticides and fungicides. Weed dry weight (kg ha^{-1}) was also included since potato farmers commonly do not use herbicides. Labour input units (h ha^{-1}) were transformed (to MJ ha^{-1}) using energetic equivalents: human labour (1.57 female and 1.96 male), animal traction (6.69) and machinery use (62.7) according to Mandal et al. (2002) and Ozkan et al. (2004). The variety maturity type used by the farmers was included: late and mid-late maturity class. Additionally, initial soil chemical characteristics were included: pH, organic carbon and total nitrogen (g kg^{-1}), extractable phosphorus and potassium (mg kg^{-1}). The statistical analysis and graphics were performed using the R statistical software (R Core Team, 2018). The classification tree was implemented using the *rpart* (Therneau and Atkinson, 2018) and *rpart.plot* (Milborrow, 2019) R-packages.

3. Results and discussion

3.1. Agronomic practices in the study zone

Table 1 shows descriptive univariate statistics of the crop management inputs and biophysical components of the potato cropping systems. Farmers' field plots area varied from 66 to 5374 m^2 ($\overline{x}$ = 1316 m^2, $\widetilde{x}$ = 954 m^2, σ_x = 1322 m^2 and CV = 1.00) in the study zone. Seed

Table 1
Descriptive univariate statistics of the crop management inputs and biophysical components of the potato cropping systems.

Biophysical component	Mean	Median	Maximum–minimum	Standard deviation	Variation coefficient[b]
Management					
Inorganic N (kg ha^{-1})	222	167	1421–0	228	1.03
Inorganic P_2O_5 (kg ha^{-1})	243	213	1343–0	221	0.91
Inorganic K_2O (kg ha^{-1})	155	134	652–0	139	0.90
Organic N (kg ha^{-1})	170	143	750–0	154	0.91
Organic P_2O_5 (kg ha^{-1})	123	97	484–0	101	0.82
Organic K_2O (kg ha^{-1})	306	242	1446–0	297	0.97
Fungicide (kg a.i. ha^{-1})[a]	0.51	0	6.61–0	1.45	2.85
Insecticide (kg a.i. ha^{-1})[a]	1.01	0.60	16.91–0	2.17	2.16
Weeds (kg ha^{-1})	2210	1178	16,412–0	2832	1.28
Human labour (MJ ha^{-1})	3160	2678	8749–1103	1580	0.50
Animal labour (MJ ha^{-1})	968	572	6532–0	1155	1.19
Machinery (MJ ha^{-1})	950	581	7446–0	1398	1.47
Main soil chemical properties					
pH	5.5	5.0	7.9–3.9	1.2	0.22
SOC (g kg^{-1})	16.3	14.5	68.0–5.0	9.7	0.60
TSN (g kg^{-1})	1.9	1.8	5–0.7	0.8	0.43
Ext. P (mg kg^{-1})	18.2	12.4	49.9–2.3	13.7	0.76
Ext. K (mg kg^{-1})	172	169	546–61	90.5	0.52

SOC: soil organic carbon; TSN: total soil nitrogen; Ext. P: extractable phosphorus (Olsen); Ext. K: extractable potassium (ammonium acetate).
[a] a.i.: active ingredient.
[b] Dimensionless.

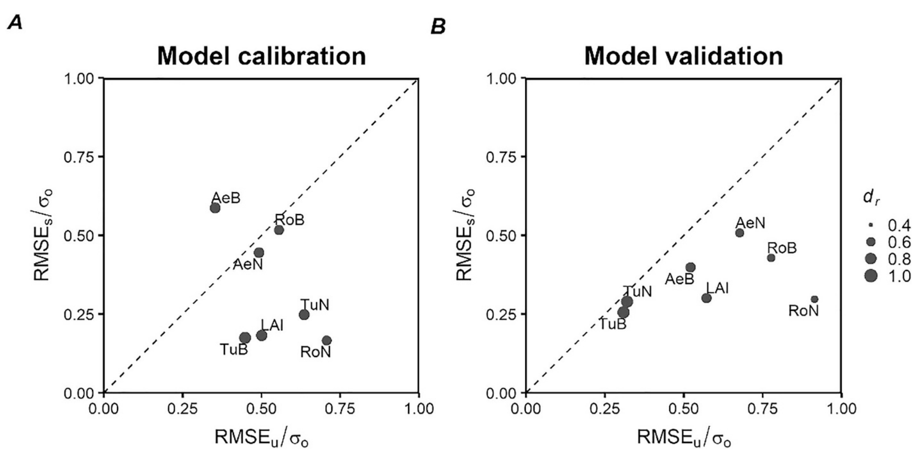

Fig. 3. Graphical representation of the model (A) calibration and (B) validation performance for the plant variables along the growing season (LAI: leaf area index, AeB: aerial biomass, TuB: tuber biomass, RoB: roots biomass, AeN: aerial N-uptake, TuN: tuber N-uptake, RoN: roots N-uptake). The root mean square error (*RMSE*) is read as the distance from the origin to the point coordinates. The coordinates represent the unsystematic *RMSE* ($RMSE_u$) and the systematic *RMSE* ($RMSE_s$) normalized by the standard deviation of the observed values (σ_O). Circles size are proportional to redefined index of agreement (d_r). Dotted line represents 1:1 reference identity line in Panel A and B.

tubers were produced or exchanged among the farmers themselves, using 1616 kg ha^{-1} seeds on average for cultivation. Potato was manually sown in October/November and harvested in April/May. *Yungay* cultivar was the most sown late-season variety, followed by the *Perricholi* cultivar (mid-late season variety) with similar growth characteristics. The use of these cultivars was analogous to other regions in the Peruvian Andes (De Valença et al., 2017; Urrea-Hernandez et al., 2016). For diminishing the infection of viruses in the seed and avoid the soil pathogens build-up, farmers usually rotated fields starting with the potato crop (e.g. barley, faba bean, wheat, and corn) and/or established fallow periods. These are practices broadly implemented in the Andes (Lutaladio et al., 2009). Most of the soils are suitable for potato (Table 1), with medium to very high soil chemical fertility (Roy et al., 2006 guidelines) and medium-textured soils. The applied fertilizer doses can vary widely (Table 1), depending on availability and adopted management practices. Organic fertilization (mainly farmyard manure) is applied at planting while inorganic fertilizer is frequently fractionated in two applications: planting and emergence (ridging). Urea, ammonium nitrate, di-ammonium phosphate, and potassium chloride are the most frequently used granulated chemical fertilizers. Not only the wide variability but also the inaccurate amount of inorganic and organic fertilizers is explained by the limited information about the potato nutritional requirements. Applied insecticides included diverse products and formulations against the most prevalent pests: *Premnotrypes* spp., *Symmetrischema tangolias* and *Liriomyza huidobrensis*. Fungicides against *Phytophthora infestans* and *Alternaria solani* are less frequently applied (Table 1). Preventive fungicides are applied only when long periods of rain occur. The duration of the cropping season depends on the agro-ecological zone (and related microclimate) where the potato is planted, and it usually enlarges as the altitude increases. Before the harvest, the shoots are cut and allowed to dry on field. In most cases, the aboveground biomass residues are incorporated into the soil during harvest. Manpower is the main labour component in the cropping systems (Table 1), mostly used during the harvesting, seeding, ridging, and weeding. In addition, animal traction is used for soil tillage, furrowing, and harvesting. Mechanization is restricted to flat fields for soil preparation, harvest transport, and postharvest activities. Most of the potato harvest is sold on field, while stored tubers represent family consumption and seed tubers for the following season. The relationship between the crop management inputs and biophysical components versus the potato yield is shown in the supplementary material (Fig. S5).

3.2. Model calibration and validation

The overall model calibration and validation are depicted in Fig. 3.

Table 2
Model calibration and validation goodness-of-fit statistics (*MAE*: mean absolute error, *RMSE*: root mean square error, $RMSE_s$: systematic *RMSE*, $RMSE_u$: unsystematic *RMSE*, d_r: redefined index of agreement).

Process	Cropping season	Observations			Summary statistics				
		Variable	n[a]	Measurement[b]	*MAE*	*RMSE*	$RMSE_s$	$RMSE_u$	d_r
Calibration	2014–2016	LAI	50	2.36 ± 2.10	0.82	1.12	0.38	1.05	0.77
		TuB	49	3428.87 ± 4645.52	1145.99	2231.92	809.55	2079.92	0.84
		AeB	50	2713.84 ± 2498.11	950.09	1710.61	1465.68	882.02	0.75
		RoB	50	419.93 ± 329.99	186.46	250.29	170.57	183.16	0.66
		TuN	49	42.73 ± 59.50	22.35	40.55	14.70	37.79	0.74
		AeN	50	77.30 ± 69.57	30.46	46.17	31.00	34.21	0.72
		RoN	50	7.93 ± 6.27	3.62	4.55	1.04	4.43	0.64
Validation	2016–2017	LAI	24	1.84 ± 1.50	0.78	0.97	0.45	0.86	0.69
		TuB	24	3930.19 ± 4182.02	915.93	1672.93	1064.69	1290.69	0.88
		AeB	24	1891.19 ± 1317.81	669.09	864.40	524.37	687.19	0.67
		RoB	24	370.91 ± 221.14	163.75	196.24	94.71	171.87	0.53
		TuN	24	64.05 ± 67.74	17.10	29.21	19.64	21.63	0.86
		AeN	23	55.70 ± 42.20	28.99	35.70	21.41	28.56	0.56
		RoN	23	6.70 ± 3.73	3.13	3.59	1.11	3.41	0.49

Plant variables along the growing season (LAI: leaf area index in m^2 leaf m^{-2} soil, AeB: aerial biomass in kg DW ha^{-1}, TuB: tuber biomass in kg DW ha^{-1}, RoB: roots biomass in kg DW ha^{-1}, AeN: aerial N-uptake in kg N ha^{-1}, TuN: tuber N-uptake in kg N ha^{-1}, RoN: roots N-uptake in kg N ha^{-1}).
[a] Number of observed-simulated pairs.
[b] Values represent mean ± standard deviation for each measurement.

Moreover, model goodness-of-fit statistics during the calibration and validation process are shown in Table 2. During the calibration process, only the aerial biomass showed systematic error, laying in the region above the 1:1 identity line (Fig. 3A). This shows the difficulty to simulate leaf and stem biomass accumulation accurately. Moreover, root biomass and root N-uptake obtained the highest *RMSE* standardized by σ_O: 0.76 and 0.73 respectively, while tuber biomass (0.48) and LAI (0.53) the lowest (Fig. 3A). Similarly, root biomass (0.66) and root N-uptake (0.64) obtained the lowest d_r while tuber biomass (0.84) and LAI (0.77) the highest (Fig. 3A and Table 2). Tuber biomass and tuber N-uptake obtained acceptable goodness-of-fit statistics for the calibration (Table 2).

All the variables laid in the unsystematic error region (below the 1:1 identity line) for the validation process (Fig. 3B). Therefore, the model errors were related to the difficulty to account for dispersion rather than due to bias in model predictions (Coucheney et al., 2015). In general, low model errors were obtained in the validation (Fig. 3B and Table 2), however d_r were predominantly lower than the calibration. Tuber biomass and tuber N-uptake were better simulated during the calibration and validation than LAI and the rest of biomass components, obtaining acceptable model fit values (Table 2) and low standardized *RMSE* (Fig. 3B). LAI, dry biomass accumulation, and N-uptake daily simulations results are provided in the supplementary material (Fig. S4).

A robust model calibration requires estimates of potential yield from high-yielding field experiments in which crops are grown without nutrient limitations or yield loss from biotic adversities (Grassini et al., 2015). The SUBSTOR-potato model has previously been reported to perform reasonably well in different environments and under diverse crop management practices (Šťastná et al., 2010; Arora et al., 2013; Raymundo et al., 2017). The LAI, biomass and N-uptake errors of the present study lie within the range of previous potato modeling studies. Specifically for tuber yield measurements, Kleinwechter et al. (2016) showed a *RMSE* of 8694 kg fresh weight (FW) ha^{-1} (~ 3524 kg dry weight (DW) ha^{-1} end of season) while Raymundo et al. (2017), 5230 kg FW ha^{-1} (~ 2120 kg DW ha^{-1} along the season). The calibration and validation results for aboveground biomass, LAI, and root dry weight variables were less accurate, obtaining *RMSE* values of 1117 kg DW ha^{-1}, 1.05 m^2 m^{-2} and 223 kg DW ha^{-1} on average, respectively. This is partly caused by the limited parameters for cultivar characterization in the SUBSTOR-potato model (Raymundo et al., 2017). Furthermore, measurement variability was large for these variables (Table 2). The systematic error for aerial DW during the calibration is linked to the aboveground biomass underestimation, a similar trend that has been found in other studies (e.g. Raymundo et al., 2017). Other potato models, such as the SPUDSIM model, also had difficulties in simulating root DW (Dathe et al., 2014). In the SUBSTOR-potato model, biomass and yield accumulation depend on LAI, light interception, and radiation use efficiency. However, non-optimal temperatures might limit actual carbon assimilation in the model. N-uptake of biomass components, representing the N-demand under potential conditions, were more difficult to model. While a general N-demand function is assumed in the SUBSTOR-potato model for each plant organ, a specific curve depending on the cultivar characteristics might improve model performance. The model does not include a phenological maturity time, which could be a potential improvement if termination of tuber growth might follow a mechanistic approach depending on cultivar (Kleinwechter et al., 2016). Similarly, the dynamic change of the specific leaf area depending on the developmental stage might improve model performance. However, a correct assessment of these parameters depends on highly detailed field measurements and further adaptations of the SUBSTOR-potato model.

3.3. On-farm yields, potential yield and yield gap

Fig. 4A shows the kernel density estimation of the actual yield (Y_a), potential yield (Y_p) and yield gap (Y_g), and Fig. 4B depicts the relationship between the yield gap closure (Y_a/Y_p) and Y_a. The actual yield DW was 7118 kg ha^{-1} on average, ranging from 710 to 18,885 kg ha^{-1} (Fig. 4A). The potential yield DW varied from 6949 to 20,216 kg ha^{-1}, with a mean value of 12,439 kg ha^{-1} (Fig. 4A). The yield gap, having a bimodal distribution, was 5321 kg DW ha^{-1} on average. The Y_g ranged from 0.1 to 95.8% of the potential yield ($\bar{x}$ = 42.1%, $\tilde{x}$ = 46.0%, σ_x = 28.14% and CV = 0.67), hence there is an important potential to increase production. The yield gap closure fluctuated from 0.04 to 0.99, with an average of 0.58 ($\tilde{x}$ = 0.54, σ_x = 0.28 and CV = 0.49) (Fig. 4B). 19% of the farmers (n = 11) obtained Y_a/Y_p ratio ~ 1, evidencing farmers with productivities close to potential. This was caused by a high use of inputs (e.g. 267 kg N ha^{-1}, 298 kg P_2O_5 ha^{-1}, 171 kg K_2O ha^{-1} of inorganic fertilizer on average) under rainfall conditions, similar to the fertilization used in the on-farm experiments. The wide range of inputs, management practices, soil nutrient features, and microclimate influence the yield variability in the study region (García, 2011).

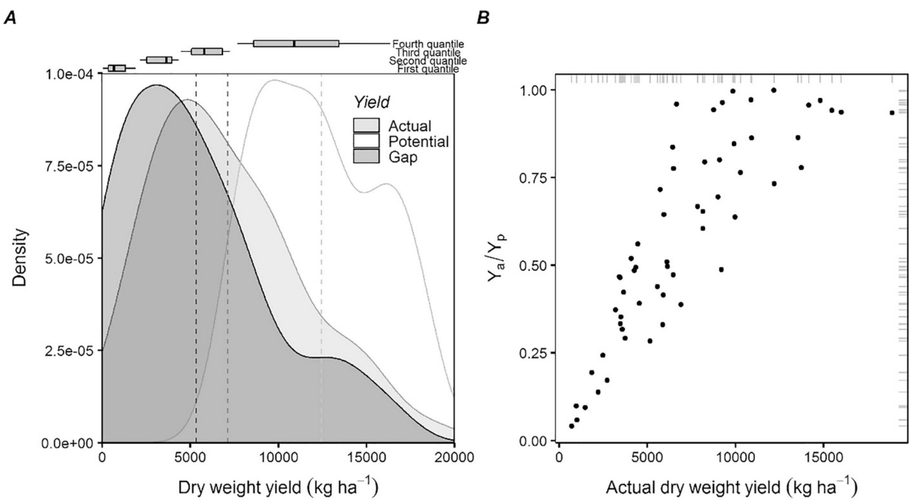

Fig. 4. (A) Kernel density estimation of actual (Y_a) and potential (Y_p) dry weight yield, and yield gap (Y_g). Dashed lines represent the mean for each variable, while the boxplots the quantiles for the Y_g in Panel A. (B) Relationship between the yield gap closure (Y_a/Y_p) and actual dry weight yield. Grey marks on axes illustrate the x- and y-values in Panel B.

The largest potato potential yield of this study was comparable with other regions. For instance, Y_p of 21–26 t DW ha^{-1} was estimated for Patagonia region in Argentina (Caldiz et al., 2002). The Y_p in China varied from 39 t FW ha^{-1} (~ 7.8 t DW ha^{-1}) in the southwest to 66 t FW ha^{-1} (~ 13.2 t DW ha^{-1}) in the north, where water supply through rainfall is close to be sufficient for potential potato growth in the southwest (Wang et al., 2018). However, the Y_p on average in the Central Peruvian Andes was lower compared to Poland (75–100 t FW ha^{-1}–15–20 t DW ha^{-1}; Verhagen et al., 2000) and the Netherlands (110 t FW ha^{-1}–22 t DW ha^{-1}; Beukema and Van der Zaag, 1990, 82 t FW ha^{-1}–16.4 t DW ha^{-1}; Kooman, 1995, 90 t FW ha^{-1}–18 t DW ha^{-1}; Haverkort and Struik, 2015). The averaged Y_a in the study region was larger than several African countries (7.5 t FW ha^{-1}–1.5 t DW ha^{-1}; Haverkort and Struik, 2015) due to better management practices and environmental conditions. Nevertheless, countries such as the Netherlands (51 t FW ha^{-1}–11 t DW ha^{-1}; Silva et al., 2017; 65 t FW ha^{-1}–13 t DW ha^{-1}; Haverkort and Struik, 2015) and the United States (120 t FW ha^{-1}–24 t DW ha^{-1}; Kunkel and Campbell, 1987) have reported higher Y_a than the Andes.

Potato yield gap has been evaluated in Chile and Argentina, obtaining an average of 52% (Haverkort et al., 2014) and 23–30% (calculated from Caldiz et al., 2001), respectively. The large Y_g of the present study (42.1%) is comparable to these studies, proving that there is still a large gap needing to be closed. While suitable environmental conditions have been reported for Andean potato agriculture to obtain high potential yields (Haverkort et al., 2014; Caldiz et al., 2002; Caldiz et al., 2001), adapted site-specific management practices might still improve the productivity in the study region. The yield gap closure (Y_a/Y_p) for potato ranges from 0.10 to 0.75 depending on the input level according to Haverkort and Struik (2015). A similar distribution but with more extreme values of the Y_a/Y_p ratio was obtained in the current study, caused by heterogeneous management practices (Table 1). Worldwide potato gap assessments have shown diverse yield gap closure values: 0.39 (Czech Republic; calculated from Šťastná et al., 2010), 0.15–0.99 (Chile; Haverkort et al., 2014), 0.10–0.37 (Zimbabwe; calculated from Svubure et al., 2015) and 0.50–0.77 (Japan; Deguchi et al., 2016).

3.4. Crop management and biophysical components of the yield gap

The classification tree for the yield gap assessment is shown in Fig. 5. The classification error was 31%, while the stratified K-fold cross-validation error was 33%. While all the variables in Table 1 were considered for the analysis, the variables actually used due to their high relative importance (between brackets) for the classification tree were: Inorganic N (8.12), Organic N (6.79), Organic P_2O_5 (5.19), Human labour (3.52) and Extractable soil phosphorus (2.71). There were 7 terminal nodes that were obtained with 10% of the number of observations approximately. Nodes were colored in grey based on the Gini impurity index (Fig. 5). The relative complex structure of the classification tree confirms the multiple factors interaction to explain the potato yield gap. Fourth and First quantiles were better classified (purer end nodes) than Third and Second quantiles (darker end nodes) (Fig. 5). Inorganic N-fertilization was used as the primary splitting node, showing its importance as main yield explaining factor (Fig. 5 and Fig. S5). The Fourth quantile (large potato yield gap) was classified based on Inorganic N (< 88 kg ha^{-1}) and Human labour (< 4196 MJ ha^{-1}), evidencing that poor inorganic N-rates with scarce human labour energy contribute to low potato productivity. On the other hand, the First quantile (small potato yield gap) was defined predominantly by the secondary splits: Inorganic N (≥ 139 kg ha^{-1}), Extractable soil phosphorus (≥ 7.3 mg kg^{-1}) and Organic N (≥ 154 kg ha^{-1}). This indicates that a low yield gap might be expected mainly with high N-inputs (inorganic and organic) when soils have medium phosphorus content. Third and Second quantiles (mid potato yield gaps) were characterized by more intricate nutrient input use, being difficult to classify (high Gini impurity index) (Fig. 5). The Third quantile was partially explained by the secondary node Inorganic N (< 139 kg ha^{-1}), while part of the Second quantile by the Extractable soil phosphorus (< 7.3 mg kg^{-1}) and Inorganic N (< 139 kg ha^{-1}).

The classification tree could be helpful as a basis to target agronomic advice and inputs to potato farmers, providing insights to close the potato yield gap. Nutrient management was identified as one of the most important yield gap explaining factors in yield gap studies worldwide (Beza et al., 2017). Similarly, the potato yield gap was primarily defined by limited use of Inorganic N in the present study (Fig. 5). Potato gap assessment showed that not only N but also P and K need to be increased to reach higher productions in China (Liu et al., 2018), while better crop management practices together with improved crop rotations are required in the Netherlands (Silva et al., 2017). However, not only N-rate but also application timing plays a relevant role because the potato crop has a shallow root system and typically low N-recovery (Palmer et al., 2013; Vos, 2009). Fractionation of chemical fertilization in three doses (best management practice) instead of two might improve productivity in the study region. Moreover, while breeding programs might be an alternative to obtain high productivities, previous research has also suggested that enhancing crop management (e.g. N-input) is more important to increase yields than the potential of potato cultivar improvement (Kleinwechter et al., 2016). Manpower (family labour) was the main labour force in smallholder potato agriculture in the study zone, especially in hillside farms. While smallholders largely depend on labor endowments (Nolte and

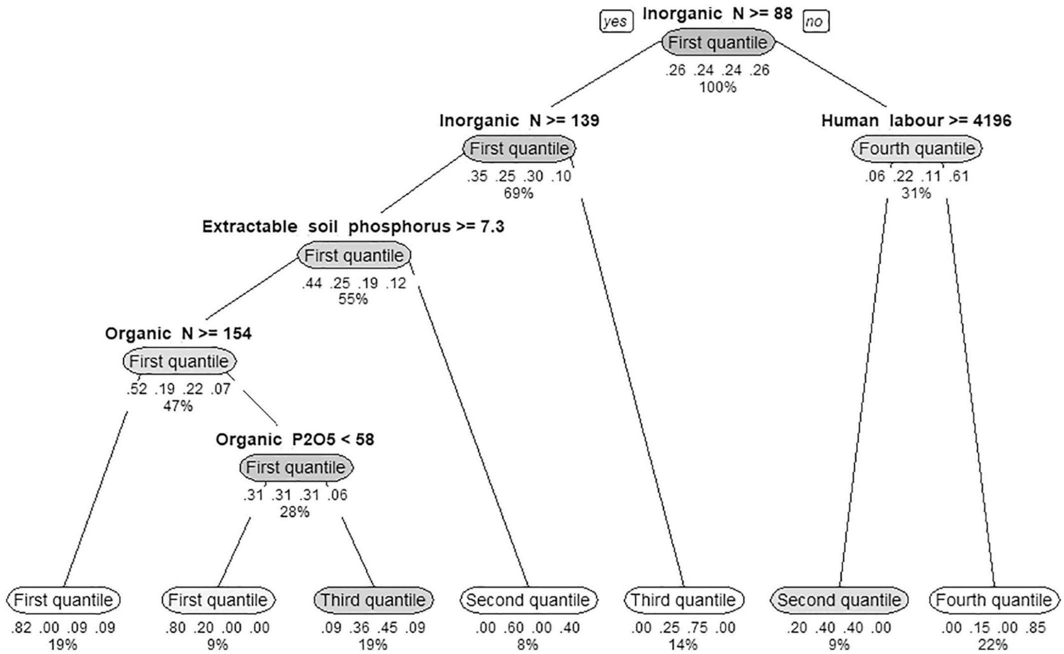

Fig. 5. Classification tree model for the potato yield gap quantiles. The grey intensity of the nodes is proportional to the Gini impurity index. Values under the nodes represent the probability of belonging to the First, Second, Third, and Fourth quantile. Node percentage value is the proportion of the data ($N = 58$) belonging to each node.

Ostermeier, 2017), limited or inefficient use can negatively affect field activities outcomes (García, 2011). Farmers with limited resources often struggle to supply sufficient labour to meet periodic labour requirements (White et al., 2005), which can explain the manpower shortage for the large potato gap group (Fourth quantile). Similarly, the self-selection of low-skilled workers into the agricultural sector in developing countries is considered to decrease agricultural productivity (Gollin et al., 2014; Lagakos and Waugh, 2013). Extractable soil phosphorus partially explained mid potato gaps (Fig. 5), however a previous study has shown that neither soil phosphorus nor potassium are strong constraining factors for the study region (García, 2011). In a similar study exploring maize yield variability under smallholder conditions, soil fertility variables were subservient to crop management variables (applying classification and regression tree modeling), being really important at low levels of resource use intensity (Tittonell et al., 2008). The use of mixed farmyard manure is a well-established practice in the study zone (best management practice), that along with inorganic inputs mainly explained the small potato gaps (First quantile) (Fig. 5). However, limited research in relation to the role of nutrient release and additional knowledge to effectively manage organic resources in Andean conditions need to be performed (Fonte et al., 2012). Terminal nodes from the classification tree can determine site-specific interventions towards closing the potato yield gap.

Bringing the potato yields to their potential would probably require more mechanization, chemical, and nutrient inputs for low-yielding farmers. These intensive land management practices can adversely affect the agro-ecosystem (Licker et al., 2010) if they are not efficiently used. Whereas farmers use several best management practices (e.g. crop rotation, manure application, and fertilizer banding), further practices can be adapted to the local conditions to enhance fertilizer recovery efficiency and increase productivity: split fertilizer application, rational amount of chemical fertilizer, and soil-based tests of nutrient supplies (Zebarth and Rosen, 2007). Similarly, integrated pest management might improve the pesticides application using a combination of physical, chemical, biological, and cultural controls to reduce crop and environmental damages (Kroschel et al., 2012). Moreover, sustainable intensification indicators and metrics for smallholder agro-ecosystems should be used as an integrated approach (Smith et al., 2017).

4. Assumptions and limitation of the study

This study combines on-farm experimental results and a crop simulation model to explore the main crop management inputs and biophysical constraints explaining the potato yield gap in the Peruvian Central Andes. However, there is still a debate of methodologies and required data from biophysical and socio-economic perspective for yield gap assessment (van Dijk et al., 2017; Rattalino Edreira et al., 2017; Guilpart et al., 2017; Beza et al., 2017; Silva et al., 2017; Grassini et al., 2015). No water limitation during the cropping seasons was assumed, because of the abundant precipitation during the cropping seasons and based on the long-term rainfall trend in the region (Sanabria et al., 2014; Sanabria and Lhomme, 2012; Silva et al., 2008). Integrated approaches considering soil processes (e.g. N-leaching) and biotic factors could expand the assessment. Nevertheless, extra soil chemical properties along the cropping seasons are required. Furthermore, modeling pests and diseases under open field conditions is still a challenge (Donatelli et al., 2017). While socio-economic features might influence the access to inputs and technology, further research integrating economical aspects can extent the results of the study. Despite these assumptions and limitations, the present study still yields valuable insights into potato cropping systems towards closing the yield gap.

5. Conclusions

This study presented quantitative analyses on potential yield and yield gaps of potato in the Central Andes of Peru. Moreover, crop management inputs and biophysical components were studied to elucidate the main managing practices explaining the potato yield gap. Performance of the SUBSTOR-potato model showed a close agreement of simulated crop biomass, tuber yield, and N-uptake with the measured data under potential conditions. Redefined index of agreement were 0.84 and 0.80 while the associated mean square error were 2232 and 916 kg ha^{-1} tuber dry weight for the calibration and validation, respectively. The average farmers' actual dry weight yield was 7118 kg ha^{-1}, however a high variability due to heterogeneous crop management was found (from 710 to 18,885 kg ha^{-1}). The potato yield gap ranged from 0.1 to 95.8% of the potential yield ($\bar{x}$ = 42.1%, $\tilde{x}$ = 46.0%, σ_x = 28.14% and CV = 0.67), hence there is an important difference that needs to be reduced. The classification tree showed that inorganic N is the main yield explaining factor. While large yield gaps (Fourth quantile) are induced by low Inorganic N (< 88 kg ha^{-1}) and scarce Human Labour energy (< 4196 MJ ha^{-1}), small yield gap (First quantile) is mainly attributed to high N-inputs (≥ 139 kg ha^{-1} inorganic and ≥ 154 kg ha^{-1} organic). Third and Second quantiles (mid potato yield gaps) were characterized by more intricate nutrient input use, being difficult to classify; the Third quantile was partially explained by Inorganic N (< 139 kg ha^{-1}), while part of the Second quantile by Extractable soil phosphorus (< 7.3 mg kg^{-1}) and Inorganic N (< 139 kg ha^{-1}). This classification can be helpful to target inputs and site-specific agronomic recommendations towards closing the potato yield gap.

The analysis suggests that there is opportunity to enhance actual potato yields in the study zone. More rational amount of inputs together with best management practices might improve potato productivity. However, sustainable potato intensification should be complemented with the expected quantification of environmental burdens under the local socio-economic constraints.

Declaration of competing interest

The authors declare no conflict of interest regarding the publication of this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.agsy.2020.102817.

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